Comparison of Distribution of Execution and Cancellation Times in Currency Market

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Abstract

The interaction between injection and annihilation (execution and cancellation) of orders can be used to isolate main characteristics of foreign exchange markets. We introduce an order book model in which orders are injected randomly and each order are either cancelled spontaneously or executed by colliding with the market price that moves randomly. This model present main lifecycle characteristics of currency markets.

Keyword: Foreign Exchange Markets, Quote Lifetime, Order Book

1. Introduction

The increase of automated traders motivates regulators to focus on lifetime of orders – amount of time spent in the market before annihilation. Recently, some researchers pays attention on order book fluctuation which includes orders not yet executed [1]. Our research focus on lifetime statistical properties of orders and construct a dynamic life cycle order book model that describes main characteristics of the market. This model will help describing how annihilation of orders affects the stability of foreign exchange markets.

2. Methodology

We use a special database of Electronic Broking System (EBS) which contains identification of every order. The database is from March 3 to March 18, 2011 and contains information about injected and annihilated orders with minimal tick time of one millisecond. Those data are used to get a better understanding of the lifecycle in foreign exchange markets.

We introduce a new model of order book in which orders are injected randomly and each order will be either cancelled spontaneously or be executed by colliding with the market price that moves randomly. This model is compared directly with the empirical market characteristics.

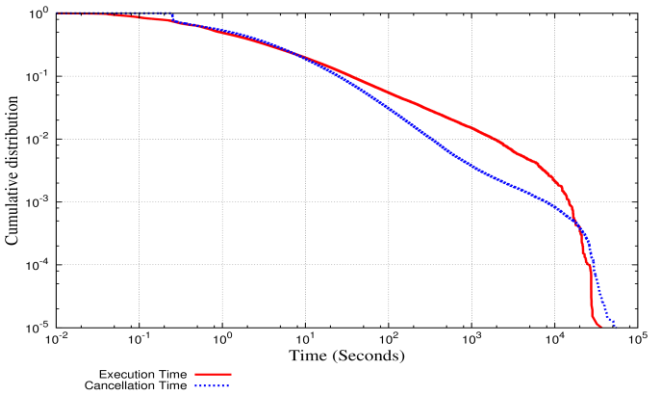


Figure 1: Log-log plot of cumulative distribution of life time of annihilated orders.

3. Conclusion

The dynamic life cycle order book model that we develop will be useful to analyze high amplitude short-time movement (such as Flash Crash) and describe impact of cancel orders on unusual high volatility episode in foreign exchange markets.

References

[1] Maslov S. (2000), Physica A 278, Issues 3-4, P. 571-578